

DELTA GRAD: TOWARDS ROBUST DEEP LEARNING VIA ADAPTIVE GRADIENT RELIABILITY

Alexandre O'Neill Mendes
Instituto Superior Técnico (IST), Lisbon, Portugal

Introduction & Motivation

Datasets frequently suffer from poor quality, including switched labels and other inconsistencies.

DeltaGrad, a novel adaptive optimizer, introduces a **Reliability metric** (R) aimed at reducing gradient variance, thereby minimizing the impact of dataset incongruence, ensuring robust generalization.

Gradient Consensus & Generalization

- **The Core Premise:** DeltaGrad relies on the signal-to-noise dynamics of gradients to measure parameter confidence.
- **Unlearned State (High Variance):** Frequent sign or magnitude shifts indicate the parameter is fitting to local batch noise rather than the global data distribution.
- **Adaptive Penalization:** By measuring gradient disagreement (Φ_j) over K steps, the optimizer reduces the update size of ungeneralized parameters.
- **Goal:** Force cautious updates until a stable, batch-agnostic direction emerges.

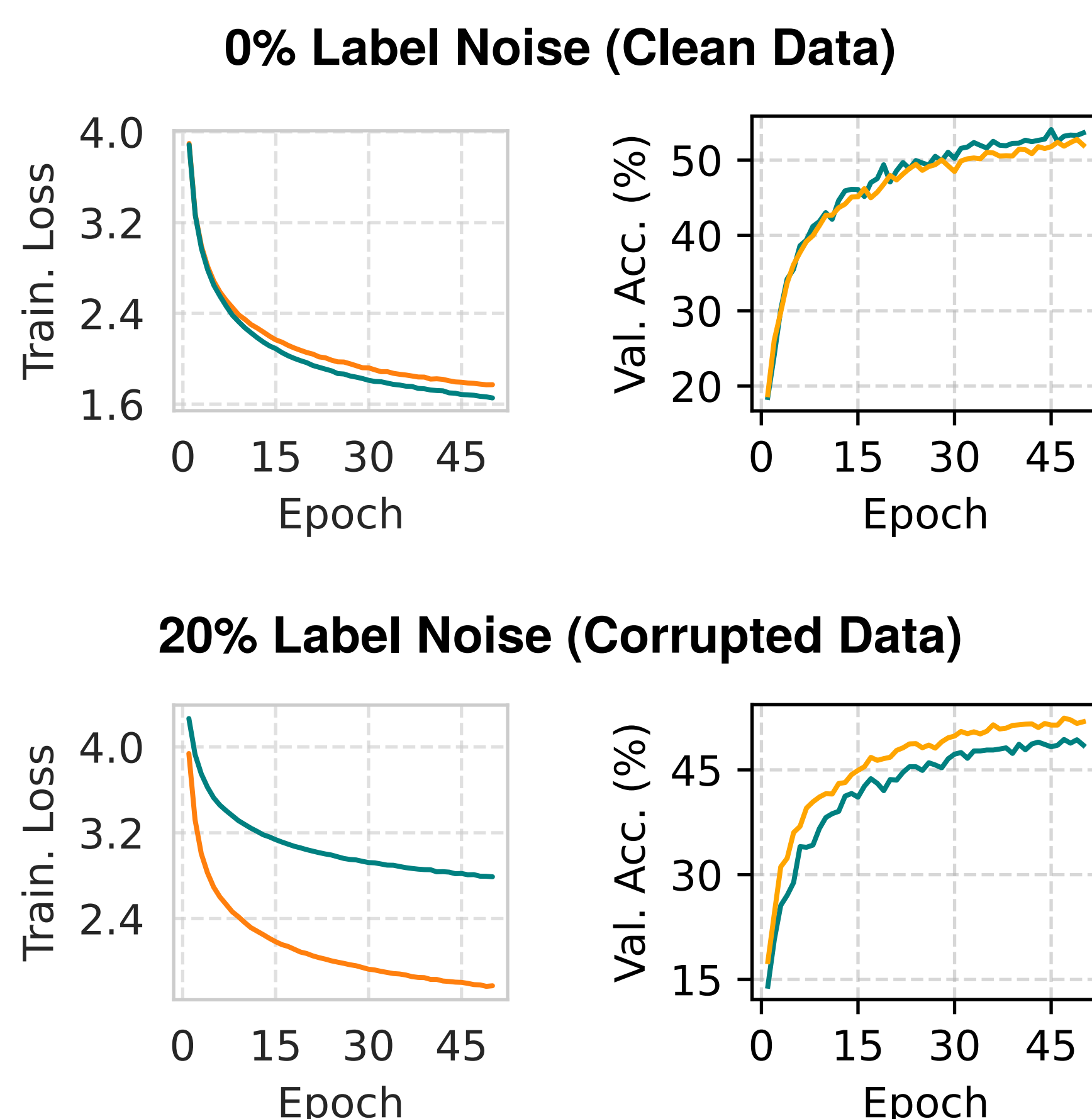
The DeltaGrad Algorithm

$$R_t = \text{clamp} \left(\frac{K - \sum_{j=0}^{K-1} \alpha^j \Phi_j}{K}, A, B \right) \quad g_s^{(t)} = \sigma g_s^{(t-1)} + (1 - \sigma) g^{(t)}$$
$$\Phi_j = \frac{|\tilde{g}^{(t)} - \tilde{g}^{(t-j)}|}{|\tilde{g}^{(t)}| + |\tilde{g}^{(t-j)}| + \epsilon} \quad \theta_{t+1} = \theta_t - \mu \cdot R_t \odot m_t$$

- where $\tilde{g}$ represents the evaluated gradient, which can be either the raw gradient (g) or the smoothed gradient (g_s), noting that $g_s = g$ when $\sigma = 0$
- where $A, B \in]0, 1.0]$ and $A \approx 0$.

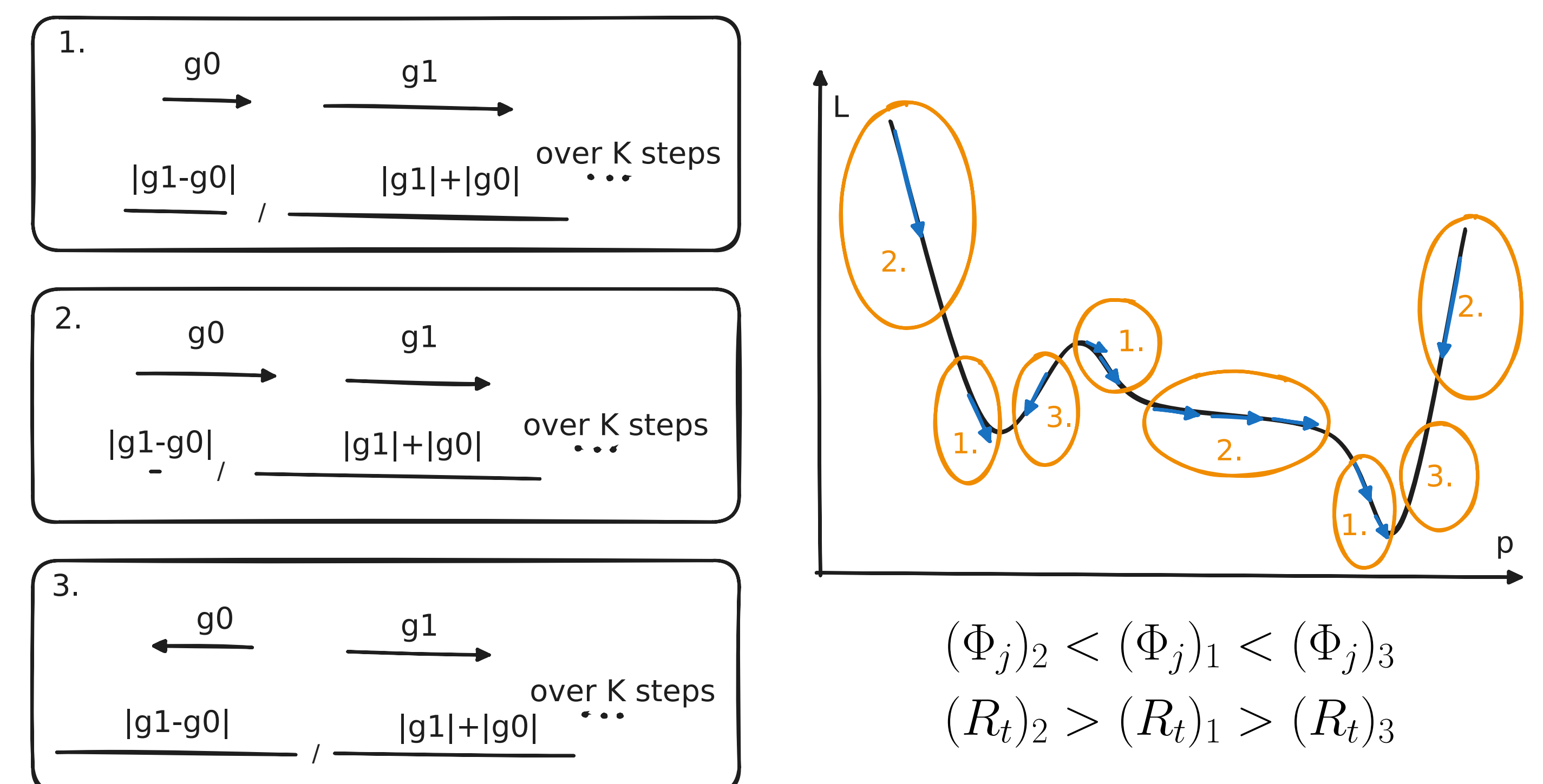
Label Noise Experiment

To evaluate **DeltaGrad**'s robustness against label noise (randomly switched labels) compared to an **Adam** baseline, models were trained using a batch size of 16, data augmentation, and regularization. The resulting training loss and validation accuracy are presented below.



- On 0% Label Noise, **DeltaGrad** actively outperforms **Adam**'s baseline accuracy.
- When noise is introduced, **Adam** blindly memorizes false labels to forcefully minimize loss.
- **DeltaGrad** detects conflicting gradients and restricts updates, plateauing at a safer loss to prevent memorization while remaining close to **Adam**'s accuracy.

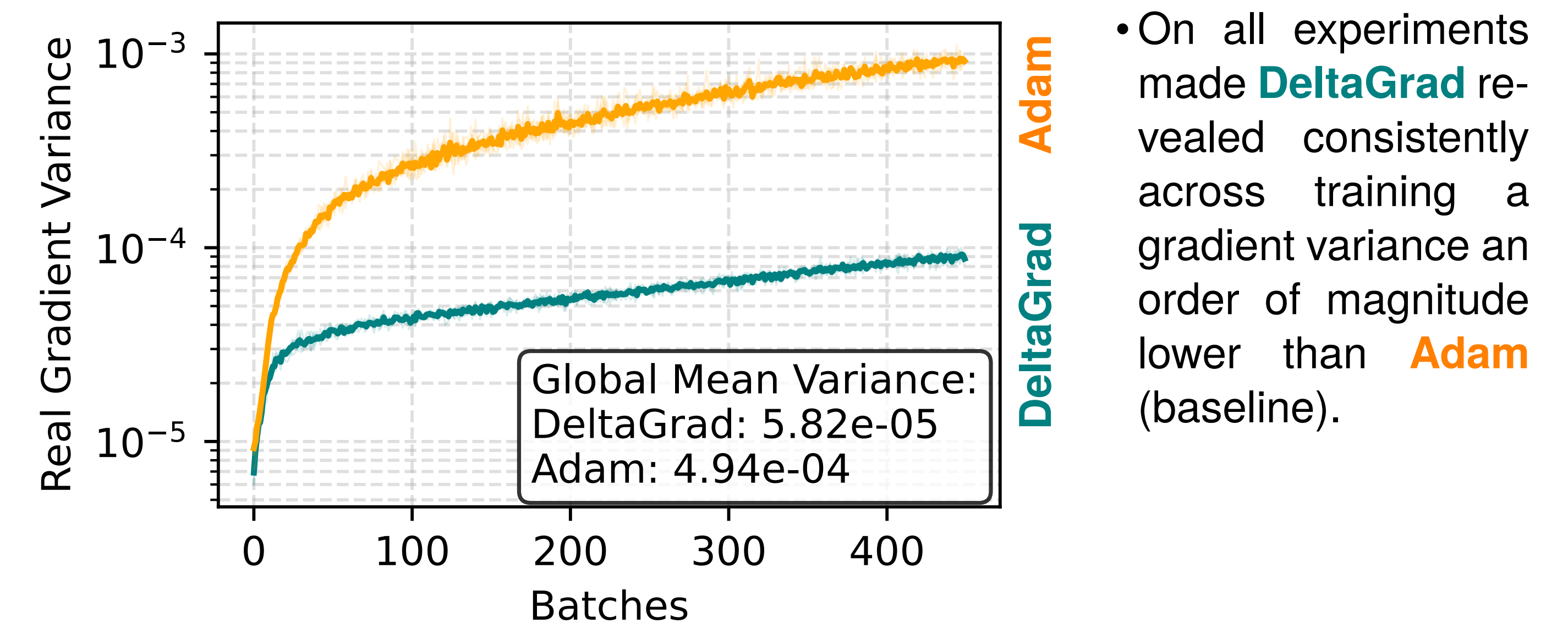
Visual Intuition of the Algorithm



- The graph above, depicting a Loss landscape with respect to a single parameter, illustrates various possible optimization behaviors.
- The marked regions are simplified examples; the optimizer's sensitivity to local topography varies significantly depending on hyperparameters like K , α , and σ .
- The clamping applied to R_t and the chosen momentum (m_t) prevent the training process from stagnating in Type 3 scenarios.

General Experimental Results

To evaluate the optimizer's mechanics in a more realistic setting, experiments (**learning rate variations and label noise**) were conducted by training a **5-layer CNN (3 Convolutional, 2 Fully Connected layers)** on the **CIFAR-100** dataset.



- On all experiments made **DeltaGrad** revealed consistently across training a gradient variance an order of magnitude lower than **Adam** (baseline).

Learning Rate Experiment

A baseline learning rate was optimized via Optuna (batch size 16) and subsequently scaled by **3x** and **10x**. Each configuration was evaluated across 5 independent runs.

The table presents the resulting accuracy standard deviation alongside the average correlation between gradient variance and R_t over these 5 runs.

LR	Acc. Std. Dev. (%)		Avg. Corr. (Grad. Var. - R_t)
	DeltaGrad	Adam	
Normal	0.71	0.64	-0.473
3x	0.93	1.76	-0.599
10x	0.93	3.18	-0.709

- **DeltaGrad** resists accuracy fluctuations significantly better than **Adam** (0.93% vs. 3.18% standard deviation).
- The strong negative correlation confirms R_t actively penalizes updates when variance spikes.

Future Work

Future work entails evaluating **DeltaGrad** across diverse architectures and tasks to fully characterize its optimization dynamics.